Gabriel Rodriguez

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EDUCATION

California State University Fullerton | Fullerton, CA — **Expected December 2026**

Bachelor of Science in Computer Science, Minor in Data Science

Citrus College | Glendora, CA — **June 2024**

Associate of Science in Computer Science

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Science & Big Data, Computational Bioinformatics

TECHNICAL SKILLS

Programming Languages: Python, SQL

Libraries/Frameworks: PyTorch, Torchvision, OpenCV, MONAI, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Concepts: Machine learning, Deep Learning, Multimodal Learning, Computer Vision, Object Oriented Programming, Feature Engineering, Prompt Engineering, Agile/Scrum

Tools: AWS (S3, SageMaker), CUDA, Git/GitHub, PyCharm, Visual Studio, Jupyter Notebook, Google Colab

PROFESSIONAL PROJECTS

Spontaneous Retinal Venous Pulsation (**SVP**) **Segmentation and Quantification** — **In Progress**

- Developing a **medical imaging research pipeline** for **semantic segmentation** of central retinal blood vessels indicative of **Spontaneous Retinal Venous Pulsation** (**SVP**) using **attention-based U-Net architectures.**
- Processing an **ophthalmic video dataset** (635 videos, 130K+ frames), applying **data augmentation, temporal modeling, and GPU-accelerated batch processing,** improving model robustness under limited labeled data.
- Extracting **quantitative imaging biomarkers** (vessel diameter, area change over time) from **segmentation** outputs to support **objective retinal health assessment**, evaluated using **Dice and IoU metrics**.

MRI Brain Tumor Classifier (**CNN**) — **November 2025**

- Designed and trained a **deep learning model** (**CNN**) for **medical image classification**, classifying brain tumors (glioma, meningioma, pituitary), achieving **89% accuracy** and **90% recall** on tumor-positive cases.
- Enhanced classification **precision** by **11%,** leveraging a **Convolutional Block Attention Module** (**CBAM**) for **spatial/channel** focus and validated model interpretability with **Grad-CAM** visualizations aligned to tumor regions.
- Minimized overfitting using **regularization** techniques incorporating **dropout**, **weight decay**, and a **learning rate scheduler** reducing validation loss by **20% on cross-validation test set**.

Diabetes Prediction Model (**ANN**) — **January 2025**

- Engineered an **Artificial Neural Network** in **PyTorch** to predict diabetes via patient **electronic medical records** (**EMR**), achieving **80% overall accuracy** and **92% recall** on diabetic cases.
- Optimized patient data preparation, applying **Pandas** and **normalization** techniques, enhancing model convergence and boosting **recall** by **8%.**
- Tuned class imbalance handling with **stratified splitting** and **class-weighted loss**, increasing **F1-score** for diabetic predictions by **10%**, ensuring high sensitivity for early diagnosis.

EXTRACURRICULAR EXPERIENCE

Data Science and Machine Learning Club | California State University Fullerton — **Present**

- **Co-authored a research paper** and **led a team of 3 students** in developing a **multimodal pipeline** for **breast cancer T-stage classification** integrating **clinical data, radiomic features**, and **3D breast MRI** (**DICOM**) **volumes**.
- Implemented a scalable **MONAI-based preprocessing and training pipeline** (**ROI extraction, voxel resampling, normalization**) achieving **97% f1-score** on T-stage 4 patients.
- Architected and trained a **late fusion 2.5D CNN with a ResNet backbone**, using stacked MRI slices to capture volumetric context and improving **T-stage 3 recall to 87%**.